

RadzenDataGrid vs. FlexKit Grid

Deep Architectural & Feature-by-Feature Parity Report

1. Executive Summary & Core Architecture

This technical report provides an exhaustive, feature-by-feature evaluation comparing **RadzenDataGrid** (from Radzen.Blazor) against **FlexKit / FlexCore GridControl** (from Fx.ControlKit.Grid). The objective is to establish complete functional, behavioral, and visual parity, ensuring FlexKit incorporates all advanced Radzen capabilities while preserving FlexKit's architectural superiority in high-latency, massive-scale data entry.

Architectural Advantage of FlexKit

While RadzenDataGrid relies strictly on standard Blazor server DOM re-rendering and basic <Virtualize> components, FlexKit employs a dedicated **Client-Buffered Typing Engine** (60fps local DOM editing with seamless Excel keyboard navigation) and **Custom Scroll Windowing** (blank-boundary guards, deferred thumb tracking, and adaptive wheel pacing) that prevent latency lag and blank flashes during intensive data operations.

2. High-Performance Scrolling & Virtualization

Feature	RadzenDataGrid	FlexKit / FlexCore Grid	Evaluation
Virtualization Engine	Uses Blazor standard <Virtualize> with VirtualizationOverscanCount.	Custom row windowing with prefix-sum pixel metrics & spacer rows.	FlexKit Superior
Scroll Track Modes	Live scroll only (continuous DOM updates).	ScrollTrack="true" (Live) or ScrollTrack="false" (Deferred).	FlexKit Superior
Blank Rows Guard	None. Rapid scrolling produces visible white blank space.	EnableScrollBoundaryGuard="true" eliminates blank row flash.	FlexKit Superior
Scroll Pacing & Slowdown	Native browser wheel only; can skip hundreds of rows uncontrollably.	EnableAdaptiveWheelScrollPacing & EnableScrollBoundarySlowdown.	FlexKit Superior
Mouse Wheel Multiplier	Not configurable.	WheelScrollScale (0.25x to 2.0x speed control).	FlexKit Superior

3. Text Editing & Data Entry Transports

Feature	RadzenDataGrid	FlexKit / FlexCore Grid	Evaluation
Typing Transport Engine	Server roundtrip (Immediate="true") or On-Blur (Immediate="false").	ClientBuffered (60fps local DOM + Excel keydown) + ServerBacked.	FlexKit Superior
Spreadsheet Batch Editing	Manual row edit states; single cell editing requires click handling.	Native EditSettings (Batch mode, single cell, multi row, formula cells).	FlexKit Superior
Keyboard Navigation	Basic browser Tab order; arrow keys don't navigate cells like Excel.	Excel-standard Arrow keys, Tab/Shift+Tab wrapping, Enter, F2, Escape.	FlexKit Superior
Row Lifecycle APIs	EditRow(), UpdateRow(), CancelEditRow(), InsertRow().	Batch-level and cell-level lifecycle methods; added row APIs.	Radzen Advantage

4. Filtering Capabilities & Query Builders

Feature	RadzenDataGrid	FlexKit / FlexCore Grid	Evaluation
Global Search Bar	None built-in (must be coded externally).	Built-in toolbar search (ShowSearchBar="true", multi-column).	FlexKit Advantage
Expression Filter	None built-in.	Built-in boolean parser ([Price] > 100 AND [Category] = 'Tools').	FlexKit Advantage
In-Header Simple Filter	FilterMode.Simple (dedicated input row below headers).	Filter Bar mode.	Radzen Advantage
Advanced Dual-Condition Menu	FilterMode.Advanced (popup with 2 operators + AND/OR).	Popup menu filter.	Radzen Advantage
Distinct Values Checklist	FilterMode.CheckBoxList (Excel-like value checklist + search).	Range filters.	Radzen Advantage
Standalone Visual Query Builder	RadzenDataFilter (recursive AND/OR group builder).	Grid-internal expression filter.	Radzen Advantage

5. Master-Detail, Hierarchy & Row Expansion

Feature	RadzenDataGrid	FlexKit / FlexCore Grid	Evaluation
Detail Row Template	<Template Context="item"> for nested sub-grids and child forms.	Supported in TreeGridControl; added to flat GridControl.	Full Parity Target
Expansion Modes	DataGridExpandMode.Single and Multiple.	Single and Multiple row expansion.	Full Parity Target
Header Expand-All Toggle	ShowExpandAll="true" in chevron column header.	Toolbar and header buttons.	Full Parity Target
Lifecycle Events & Methods	RowExpand, RowCollapse, ExpandRow(), CollapseRow().	RowExpand, RowCollapse, ExpandAll(), CollapseAll().	Full Parity Target

6. Sorting, Multi-Column Priority & Comparers

Feature	RadzenDataGrid	FlexKit / FlexCore Grid	Evaluation
Single Column Sort	AllowSorting="true" (Ascending, Descending, None).	AllowSorting="true" with sort direction toggles.	Full Parity
Multi-Column Sorting	AllowMultiColumnSorting="true" via Shift+Click.	AllowMultiSorting="true".	Full Parity
Sort Priority Number Badges	ShowMultiColumnSortingIndex="true" (shows 1, 2, 3 on headers).	Visual directional arrows; index badge support.	Full Parity Target
Custom Comparers	SortComparer and SortProperty.	LINQ lambda comparers and custom accessors.	Full Parity

7. Grouping, Aggregates & Group Footers

Feature	RadzenDataGrid	FlexKit / FlexCore Grid	Evaluation
Drag-and-Drop Grouping Bar	AllowGrouping="true", HideGroupedColumn.	AllowGrouping="true" with group drop rail.	Full Parity
Group Header Templates	GroupHeaderTemplate.	Group header renderers with VB6 +/- icons.	Full Parity
Group Footer Summaries	GroupFooterTemplate, GroupFootersAlwaysVisible.	Table-wide summary footer; group footer aggregates.	Full Parity Target
Aggregate Calculations	Sum, Average, Count, Min, Max per group level.	Sum, Average, Count, Min, Max.	Full Parity

8. Column Operations & Selection Models

Feature	RadzenDataGrid	FlexKit / FlexCore Grid	Evaluation
Pinned / Frozen Columns	Left and Right column pinning (Frozen, FrozenPosition).	Left and Right column pinning (IsFrozen, FrozenPosition).	Full Parity
Column Resizing & Reordering	Interactive drag resize with auto-fit; drag header reorder.	Interactive drag resize with auto-fit; drag header reorder.	Full Parity
Column Chooser / Picker	AllowColumnPicking="true" dropdown checklist.	ShowColumnChooser="true" dialog and rail picker.	Full Parity
Composite / Multi-Headers	Nested <Columns> inside RadzenDataGridColumn.	Grouped super-headers & TreeGridColumn.	Full Parity
Row & Checkbox Selection	Single/Multiple row selection; Checkbox column with Select All.	Single/Multiple row selection; Checkbox column with Select All.	Full Parity
Excel Cell Range Drag Selection	Cell selection mode only.	Drag-select rectangular cell regions across multiple columns/rows.	FlexKit Superior

9. Specialized Grid Controls in Ecosystem

Component	Radzen Ecosystem	FlexKit / FlexCore Equivalent
DropDown DataGrid	RadzenDropDownDataGrid (multi-column searchable combo popup).	DropDownGridControl (combines dropdown search with embedded GridControl).
Pivot Grid Analysis	RadzenPivotDataGrid (cross-tab dimensional aggregation).	PivotControl (full pivot grid with row/column fields and sub-totals).
Hierarchical Tree Grid	RadzenTreeGrid (self-referencing tree nodes).	TreeGridControl (hierarchical tree grid with indentation and level metrics).

Component	Radzen Ecosystem	FlexKit / FlexCore Equivalent
Visual Query Builder	RadzenDataFilter (recursive rule builder with AND/OR trees).	DataFilterControl (standalone dynamic boolean filter generator).

10. Action Plan for 100% Feature Parity

To ensure FlexKit / FlexCore Grid is definitively feature-complete against RadzenDataGrid while retaining its industry-leading performance architecture, the following additions are targeted:

- In-Header & Popup Filter Modes:** Add GridFilterMode supporting Simple (in-header inputs), Advanced (2-condition modal with AND/OR), and CheckBoxLayout (distinct values scanner).
- Master-Detail Row Expansion:** Introduce DetailTemplate, ExpandMode, ShowExpandColumn, ShowExpandAll, and RowExpand/RowCollapse to GridControl.
- Multi-Column Sort Number Badges:** Display priority index numbers (1, 2, 3) on sorted column headers when multi-sorting.
- Group Footer Aggregates:** Support GroupFooterTemplate with per-group aggregate summaries.
- Row-Level Editing APIs:** Provide EditRow(), SaveRow(), CancelEditRow(), and InsertRow() alongside cell batch editing.
- Complementary Components:** Add DropDownGridControl and DataFilterControl to the FlexCore control kit.